

Siyu Chen

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Research Interest

Deep Learning, Foundation Model, Various Aspects of Statistics, Reinforcement Learning

Education

Doctoral Student	Yale University , Statistics and Data Science • <i>Advisor</i> : Prof. Zhuoran Yang	Sep 2022 to now
Bachelor of Science	Tsinghua University , Electronic Engineering	Sep 2018 to May 2022

Experience

Yale University , Teaching Assistant for Information Theory	CT, USA Feb 2024 to May 2024
Google Research , Student researcher on LLM test-time scaling and agentic system	CT, USA June 2025 to Dec 2025

Publications

- Altabaa, A., S. Chen, J. Lafferty, and Z. Yang (2025). “Unlocking Out-of-Distribution Generalization in Transformers via Recursive Latent Space Reasoning”. In: *arXiv preprint arXiv:2510.14095*.
- Chen, S., H. Sheen, T. Wang, and Z. Yang (2024). “Unveiling Induction Heads: Provable Training Dynamics and Feature Learning in Transformers”. In: *Advances in Neural Information Processing Systems*. Vol. 37, pp. 66479–66567.
- Chen, S., H. Sheen, X. Xiong, T. Wang, and Z. Yang (2025). “Taming Polysematicity in LLMs: Provable Feature Recovery via Sparse Autoencoders”. In: *arXiv preprint arXiv:2506.14002*.
- Chen, S., M. Wang, and Z. Yang (2023). “Actions Speak What You Want: Provably Sample-Efficient Reinforcement Learning of the Quantal Stackelberg Equilibrium from Strategic Feedbacks”. In: *arXiv preprint arXiv:2307.14085*.
- Chen, S., Y. Wang, Z. Wang, and Z. Yang (2023). “A unified framework of policy learning for contextual bandit with confounding bias and missing observations”. In: *arXiv preprint arXiv:2303.11187*.
- Chen, S., B. Wu, M. Lu, Z. Yang, and T. Wang (2025). “Can Neural Networks Achieve Optimal Computational-statistical Tradeoff? An Analysis on Single-Index Model”. In: *The Thirteenth International Conference on Learning Representations, (oral)*.
- Chen, S., J. Wu, Y. Wu, and Z. Yang (2023). “Learning to incentivize information acquisition: Proper scoring rules meet principal-agent model”. In: *International Conference on Machine Learning*. PMLR, pp. 5194–5218.
- Chen, S., D. Yang, J. Li, S. Wang, Z. Yang, and Z. Wang (2022). “Adaptive model design for Markov decision process”. In: *International Conference on Machine Learning*. PMLR, pp. 3679–3700.
- He, J., S. Chen, F. Zhang, and Z. Yang (2024). “From Words to Actions: Unveiling the Theoretical Underpinnings of LLM-Driven Autonomous Systems”. In: *International Conference on Machine Learning*. PMLR, pp. 17807–17841.
- He, J., X. Pan, S. Chen, and Z. Yang (n.d.). “In-Context Linear Regression Demystified: Training Dynamics and Mechanistic Interpretability of Multi-Head Softmax Attention”. In: *Forty-second International Conference on Machine Learning*.

- Hu, X., F. Zhang, S. Chen, and Z. Yang (2024). “Unveiling the Statistical Foundations of Chain-of-Thought Prompting Methods”. In: *arXiv preprint arXiv:2408.14511*.
- Sheen, H., S. Chen, T. Wang, and H. H. Zhou (2024). “Implicit Regularization of Gradient Flow on One-Layer Softmax Attention”. In: *arXiv preprint arXiv:2403.08699*.
- Siyu, C., S. Heejune, W. Tianhao, and Y. Zhuoran (2024). “Training Dynamics of Multi-Head Softmax Attention for In-Context Learning: Emergence, Convergence, and Optimality”. In: *The Thirty Seventh Annual Conference on Learning Theory*. PMLR, pp. 4573–4573.
- Siyu Chen Theodor Misiakiewicz, I. Z. and P. Zhang (2025). “An Optimized Franz-Parisi Criterion and its Equivalence with SQ Lower Bounds”. In: *The Thirty-Ninth Annual Conference on Neural Information Processing Systems*, **(oral)**.
- Wu, J., S. Chen, M. Wang, H. Wang, and H. Xu (2024). “Contractual Reinforcement Learning: Pulling Arms with Invisible Hands”. In: *arXiv preprint arXiv:2407.01458*.
- Zhang^{*}, Y., S. Chen^{*}, Z. Yang, M. Jordan, and Z. Wang (2021). “Wasserstein flow meets replicator dynamics: A mean-field analysis of representation learning in actor-critic”. In: *Advances in Neural Information Processing Systems* 34, pp. 15993–16006.

Skills

Languages: C++, C, Python, R ; **ML Libraries:** Pytorch, triton

Packages: Pytorch